LOGARITHMIC AND EXPONENTIAL EQUATIONS

Which method?

Where the unknown sits decides the family; the form of the equation decides the method.

START

Where is the unknown?

inside a log

What form? **one log = k**

Convert to exponential form

$\log_a(x) = k$ gives $x = a^k$

→ Basic form

$\log_2(x) = 5 \rightarrow x = 32$

What form? **log = log**

One-to-one property

$\log_a(M) = \log_a(N)$ gives $M = N$

→ Logs on both sides

$\log_4(3x + 2) = \log_4(x + 10) \rightarrow x = 4$

What form? **sum or difference**

Condense, then convert

combine with the logarithm rules into one log

→ Combining logarithms

$\log_2(x) + \log_2(x - 2) = 3 \rightarrow x = 4$

What form? **quadratic in log**

Substitute $u = \log_a(x)$

solve the quadratic in u, then back-substitute

→ Substitution

$(\log_2 x)^2 - 5\log_2 x + 6 = 0 \rightarrow x = 4$ or 8

in an exponent

Exponentials on both sides? **no**

Take a log of both sides

$a^x = B$: take log or ln, then divide

→ Exponential equations

$3^x = 7 \rightarrow x = \log(7)/\log(3) \approx 1.771$

Exponentials on both sides? **yes**

Take logs, collect x

expand both sides and solve the linear equation in x

→ Exponentials on both sides

$2^{x+3} = 5^{x-1} \rightarrow x \approx 4.026$

LOG SIDE Check every answer against the domain: the argument of each logarithm must be positive.